Dr. Yang Xiao

Assistant Professor of Chemical Engineering
Principal Investigator of *Reaction Engineering and Catalysis Science Laboratory (RECSL)*
Department of Chemical Engineering / Institute for Micromanufacturing (IfM)
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Education

Postdoc	**Purdue University** (West Lafayette, Indiana)	2015 - 2018
Ph.D. Chemical Engineering	**Southeast University** (Nanjing, China)	2010 - 2015
Co-advisors:		
Prof. Arvind Varma, Purdue University		*2012 - 2015*
Prof. Guomin Xiao, Southeast University		*2010 - 2012*
B.S. Chemical Engineering	**Southeast University** (Nanjing, China)	2006 - 2010

Ph. D. Dissertation

Yang Xiao, Heterogeneous Catalysis for Biodiesel Production and Utilization of its Byproduct Crude Glycerol by Selective Oxidation, Southeast University (China), May **2015**, 137 + xii pages. Co-advised by Prof. Arvind Varma *(Purdue University)* and Prof. Guomin Xiao *(Southeast University)*

Appointments

Assistant Professor	Louisiana Tech University	2022 - Present
Senior Research Engineer (PI)	Purdue University	2019 - 2022
Research Scientist	Purdue University	2018 - 2019

Grants and Awards

LAMDA Track 1B Award	LAMDA/NSF EPSCoR	2023-present
Research Enhancement Award (REA)	LaSPACE/NASA EPSCoR	2023-present
Young Researcher Travel Award	ISCRE27	2023
IFM Research Innovation Initiative (RII)	Louisiana Tech University	2022-present
COES RED Travel Grant	Louisiana Tech University	2023
COES Start-up Fund	Louisiana Tech University	2022-present

Research Group

Ph.D. Graduate Students:

Tobias K. Misicko	Louisiana Tech University	*2022 - present*
Xiaoyang Gao *(co-advised)*	Louisiana Tech University	*2022 - present*
Mst. Nusrat Yasmin	Louisiana Tech University	*2023 - present*

Undergraduate Students:

Brady Duplessis	Louisiana Tech University	*2022 - 2023*
Paul Macip	Louisiana Tech University	*2022 - 2023*
Jacob Christ	Louisiana Tech University	*2022 - present*
Sarah Siharath	Louisiana Tech University	*2023 - present*
Robert Martin	Louisiana Tech University	*2023 - present*

High School Students:

Piper Smith		*2023*
Mir Z. Ali		*2023*

Teaching Experience

Sustainable Chemical Processes (CMEN 450)	Louisiana Tech University	Present (Fall Quarter)
Sustainable Chemical Processes (CMEN 557)	Louisiana Tech University	Present (Fall Quarter)
Transport Phenomena (CMEN 304)	Louisiana Tech University	Present (Spring Quarter)
Engineering Problem Solving II (ENG 121)	Louisiana Tech University	Present (Winter Quarter)
Heat and Mass Transfer (CHE 378)	Purdue University	2021
Chemical Reaction Engineering (CHE 660) **(Co-Instructor)**	Purdue University	2019

Research Publications

$*$ *corresponding authors;* † *these authors contributed equally*

30. Zhe Li†, Tobias K. Misicko†, Fan Yang†, Xiaopeng Liu, Zhenwei Wu, Xiaoyang Gao, Tao Ma, Jeffrey T. Miller, Daniela S. Mainardi, Collin D. Wick, Zhenhua Zeng*, Yang Xiao*, Yue Wu*, Nano Research, 2023, In Press, DOI: 10.1007/s12274-023-6022-2.
29. Zhichao Shang, Teng Wang, Aoxia Ren, Yong Yu, Yan Zheng, Yuan Tao, Peizhong Feng, Yang Xiao*, Xiaohong Wang*, Hollow Macroporous CeO_2/β-Bi_2O_3 Heterostructure Sphere via One-step Spray Solution Combustion Synthesis for Efficient Photocatalysis, Applied Surface Science, **2023**, 619, 156718.
28. Yang Xiao*, Anand Ramanathan, Bala Subramaniam and Arvind Varma, Guaiacol Hydrodeoxygenation and Hydrogenation over Bimetallic Pt-M (Nb, W, Zr)/KIT-6 Catalysts with Tunable Acidity, ACS Sustainable Chemistry & Engineering, **2022**, 10 (15), 4831-4833.
27. Zhichao Shang, Yong Yu, Hang Yang, Zhongxiang Yang, Yang Xiao*, Xiaohong Wang* One-step Solution Combustion Synthesis of Micro/nano-scale Porous Cu/CeO_2 with Enhanced Photocatalytic Properties, Journal of Rare Earths, **2023**, 41 (2), 250-258.
26. Zhe Li†, Yang Xiao*†, Prabudhya Roy Chowdhury, Zhenwei Wu, Tao Ma, Johnny ZhuChen, Gang Wan, Tae-Hoon Kim, Dapeng Jing, Peilei He, Pratik J. Potdar, Lin Zhou, Zhenhua Zeng, Xiulin Ruan, Jeffrey T. Miller, Jeffrey P. Greeley, Yue Wu*, Direct Methane Activation by Atomically Dispersed Platinum Nanolayers on Two-dimensional Metal Carbides (MXenes), Nature Catalysis, **2021**, 4, 882-891.
25. Ruiping Wei, Xumin Qu, Yang Xiao*, Jingdeng Fan, Gaoli Geng, Lijing Gao, Guomin Xiao, Glycerol Hydrogenolysis Propanediols over Silicotungstic Acid Catalysts Intercalated with CuZnFe Hydrotalcite-

like Compounds **2021**, Catalysis Today, **2021**, 368, 224-231.

24. Chunxiao Xu, Yayong Li, Ryan A. Adams, Vilas G. Pol, Yang Xiao, Arvind Varma, Pengwan Chen, One-Step Combustion Synthesis of Carbon-Coated NiO/Ni Composites for Lithium and Sodium Storage, Journal of Alloys and Compounds, **2021**, 310, 110578.
23. Yuanfeng Wu, Yang Xiao, Hui Yuan, Zongqi Zhang, Shengbin Shi, Ruiping Wei, Lijing Gao and Guomin Xiao, Imidazolium ionic liquid functionalized UiO-66-NH_2 as highly efficient catalysts for chemical fixation of CO_2 into cyclic carbonates, *Microporous and Mesoporous Materials*, **2021**, 310, 110578.
22. Feng Jiang, Shanshan Wang, Bing Liu, Jie Liu, Li Wang, Yang Xiao, Yuebing Xu, and Xiaohao Liu, Insights into the Influence of CeO_2 Crystal Facet on CO_2 Hydrogenation to Methanol over Pd/CeO_2 Catalysts *ACS Catalysis*, **2020**, 10 (19), 11493–11509.
22. Yuanfeng Wu, Yang Xiao, Hui Yuan, Zongqi Zhang, Shengbin Shi, Ruiping Wei, Lijing Gao and Guomin Xiao, Imidazolium ionic liquid functionalized UiO-66-NH_2 as highly efficient catalysts for chemical fixation of CO_2 into cyclic carbonates, *Microporous and Mesoporous Materials*, **2021**, 310, 110578.
21. Zhichao Shang, Zhongxiang Yang, Yang Xiao*, and Xiaohong Wang*, Ordered mesoporous Ag/CeO_2 nanocrystalline via silica-templated solution combustion for enhanced photocatalytic performance, *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, **2020**, 604, 125301.
20. Yang Xiao*, Rexonni Lagare, Lindsey Blanshan, Enrico N. Martinez, and Arvind Varma, Refinement of the Kinetic Model for Guaiacol Hydrodeoxygenation over Platinum Catalysts, *AIChE Journal*, **2020**, 66 (5), e16913.
19. Kanak Roy, Luca Artiglia, Yang Xiao, Arvind Varma and Jeroen A. van Bokhoven, Role of Bismuth in the Stability of Pt-Bi Bimetallic Catalyst for Methane Mediated Deoxygenation of Guaiacol, an APXPS study, *ACS Catalysis*, **2019**, 9 (4), 3694-3699.
18. Johnny ZhuChen, Zhenwei Wu, Xiaoben Zhang, Slgi Choi, Yang Xiao, Arvind Varma, Wei Liu, Guanghui Zhang and Jeffrey T. Miller, Identification of the Structure of the Bi Promoted Pt Non-oxidative Coupling of Methane Catalyst: A Nanoscale Pt_3Bi Intermetallic Alloy, *Catalysis Science & Technology*, **2019**, 9, 1349-1356.
17. Yuan Wang, Yang Xiao* and Guomin Xiao, Sustainable Value-added C_3 Chemicals from Glycerol Transformations: A Mini Review for Heterogeneous Catalytic Processes, **2019**, *Chinese Journal of Chemical Engineering*, **2019**, 27, 1536-1542.
16. Xiaoni Du, Xiaohong Wang and Yang Xiao, $Bi_5O_7NO_3$ and Ag/$Bi_5O_7NO_3$ Composites: One-step Solution Combustion Synthesis, Characterization and Photocatalytic Properties, *CrystEngComm*, **2018**, 20, 7536-7542.
15. Yang Xiao, Yuan Wang and Arvind Varma, Low-Temperature Selective Oxidation of Methanol over Pt-Bi Bimetallic Catalysts, *Journal of Catalysis*, **2018**, 363, 144-153.
14. Yang Xiao and Arvind Varma, Highly Selective Nonoxidative Coupling of Methane over Pt-Bi Bimetallic Catalysts, *ACS Catalysis*, **2018**, 8 (4), 2735-2740.
13. Yang Xiao and Arvind Varma, Bio-oil Upgrading Using Methane: A Mechanistic Study of Model Com-

pound Guaiacol Reactions over Pt-Bi Bimetallic Catalysts, *ACS Sustainable Chemistry & Engineering*, **2018**, 6 (12), 17368-17375.

12. Yang Xiao and Arvind Varma, Kinetics of Glycerol Conversion to Hydrocarbon Fuels over Pd/H-ZSM-5 Catalyst, *AIChE Journal*, **2017**, 63 (12), 5445-5451.
11. Yang Xiao and Arvind Varma, Kinetics of Guaiacol Deoxygenation over the Pt-Bi Catalyst, *Reaction Chemistry & Engineering*, **2017**, 2, 36-43.
10. Yang Xiao, Jeffrey Greeley, Arvind Varma, Zhi-Jian Zhao and Guomin Xiao, An Experimental and Theoretical Study of Glycerol Oxidation to 1,3-Dihydroxyacetone Over Bimetallic Pt-Bi Catalysts, *AIChE Journal*, **2017**, 63 (2), 705-715. **(as Most Read Article)**
9. Yang Xiao and Arvind Varma, Conversion of Glycerol to Hydrocarbon Fuels via Bifunctional Catalysts, *ACS Energy Letters*, **2016** 1 (5), 963-968.
8. Danni Gao, Yang Xiao and Arvind Varma, Guaiacol Hydrodeoxygenation over Platinum Catalyst: Reaction Pathways and Kinetics, *Industrial & Engineering Chemistry Research*, **2015**, 54 (43), 10638-10644.
7. Yang Xiao and Arvind Varma, Catalytic Deoxygenation of Guaiacol Using Methane, *ACS Sustainable Chemistry & Engineering*, **2015**, 3 (11), 2606-2610.
6. Haoyang Li, Xiaomei Pan, Yang Xiao, Guomin Xiao and Jinjin Huang, Simulation of Biodiesel Industrial Production via Solid Base Catalyst in a Fixed-bed Reactor, *Journal of Southeast University (English Edition)*, **2014**, 30 (3), 380-386.
5. Yang Xiao, Guomin Xiao and Arvind Varma, A Universal Procedure for Crude Glycerol Purification from Different Feedstocks in Biodiesel Production: Experimental and Simulation Study, *Industrial & Engineering Chemistry Research*, **2013**, 52 (39), 14291-14296.
4. Yang Xiao, Haoyang Li, Guomin Xiao, Lijing Gao and Xiaomei Pan, Simulation of the Catalytic Reactive Distillation Process for Biodiesel Production via Transesterification, *International Conference on Materials for Renewable Energy and Environment*, Chengdu, China, **2013**, 1, 196-199.
3. Guomin Xiao, Haoyang Li and Yang Xiao, Applications of Chemical Simulation Softwares in the Course of Transport Phenomena, *Chemical Industry Times*, **2012**, 26 (10), 53-59.
2. Yang Xiao, Lijing Gao, Guomin Xiao, Baosong Fu and Lei Niu, Experimental and Modeling Study of Continuous Catalytic Transesterification to Biodiesel in a Bench-Scale Fixed-Bed Reactor, *Industrial & Engineering Chemistry Research*, **2012**, 51 (37), 11860-11865.
1. Yang Xiao, Lijing Gao, Guomin Xiao and Jianhua Lv, Kinetics of the Transesterification Reaction Catalyzed by Solid Base in a Fixed-bed Reactor, *Energy & Fuels*, **2010**, 24 (11), 5829-5833.

U. S. Patents and Patent Applications

6. Yang Xiao, Arvind Varma, Zhe Li and Yue Wu, Transition Metal Carbides for Catalytic Methane Activation, Patent No. US11524279B1, December 13, **2022**.
5. Yang Xiao, Arvind Varma, Anand Ramanathan, and Bala Subramaniam, Method of Enhanced Aromatic Products from Bio-oil Upgrading , Patent Application US20220356137A1, June 22, **2019**.
4. Yang Xiao and Arvind Varma, Catalytic Deoxygenation of Bio-Oils Using Methane, Patent, No. US10023809

B2, July 17, **2018**.

3. Yang Xiao and Arvind Varma, Method of Producing Formaldehyde from Methanol, Patent No. US10392333B2, December 19, **2018**.
2. Yang Xiao and Arvind Varma, Method of Conversion of Glycerol to Hydrocarbon Fuels, Patent No. US10781376B2, October 3, **2018**.
1. Yang Xiao and Arvind Varma, Non-oxidative Production of Hydrocarbon from Methane, Patent No. US10450247B2, September 19, **2018**.

Conference Presentations

37. Yang Xiao, Zhe Li, Tobias Misicko, Jeff T. Miller, Yue Wu, Arvind Varma Atomically Thin Platinum Nanolayers on MXene for Catalytic Non-oxidative Coupling of Methane, *The 27th International Symposium for Chemical Reaction Engineering (ISCRE 27)*, Quebec City, Quebec, Canada, June 11-14, 2023.
36. Tobias Misicko and Yang Xiao Thermal Runaway of Catalytic Reactors: an Experimental,Modeling and Machine-Learning Study *Southeastern Catalysis Society 2023 Annual Symposium*, Tuscaloosa, AL, February 27 - 28, 2023.
35. Yang Xiao, Tobias Misicko, Jeffrey Miller and Yue Wu, Atomically Thin Pt Nanolayer Catalysts Supported on Two-dimensional Molybdenum Titanium Carbide (MXene) for Shale Gas Conversion, *Southeastern Catalysis Society 2023 Annual Symposium*, Tuscaloosa, AL, February 27 - 28, 2023.
34. Zhe Li, Yang Xiao, Prabudhya Chowdhury, Zhenwei Wu, Tao Ma, Johnny Zhuchen, Gang Wan, Tae-Hoon Kim, Dapeng Jing, Peilei He, Pratik Potdar, Lin Zhou, Zhenhua Zeng, Xiulin Ruan, Jeffrey T. Miller, Jeffrey Greeley, Yue Wu and Arvind Varma, Nonoxidation Coupling of Methane over Nano-Layer Platinum Catalysts on Two-Dimensional Metal Carbides (MXenes), *AIChE Annual Meeting*, Phoenix, AZ, November 13 - 18, 2022.
33. Yang Xiao, Anand Ramanathan, Bala Subramaniam and Arvind Varma, Guaiacol Hydrodeoxygenation and Hydrogenation over Bimetallic Pt-M (Nb, W, Zr)/KIT-6 Catalysts with Tunable Acidity *AIChE Annual Meeting*, Phoenix, AZ, November 13 - 18, 2022.
32. Yang Xiao, Pratik Potdar, Kaida Liu, Arvind Varma and Guomin Xiao A Machine Learning Tool for Thermal Runaway Prediction of Chemical Reactors *AIChE Annual Meeting*, Virtual, November 16 - 20, 2020.
31. Yang Xiao, Feng Jiang, Guomin Xiao and Arvind Varma, CO_2 Hydrogenation to Ethanol over Pd/Bi_2O_3 Catalysts: The Synergistic Effect of Pd Particle Size and Surface Oxygen Vacancy *AIChE Annual Meeting*, Virtual, November 16 - 20, 2020.
30. Yang Xiao and Arvind Varma, Parametric Sensitivity and Runaway in Fixed-Bed Reactors: Example of Methanol Selective Oxidation over Pt-Bi Catalysts, *AIChE Annual Meeting*, Orlando, Florida, November 10-15, **2019**.
29. Yang Xiao, Anand Ramanathan, Bala Subramaniam and Arvind Varma, Enhanced Aromatic Selectivity during Deoxygenation of Phenolic Model Compounds over Bifunctional Catalysts, *AIChE Annual Meeting*, Orlando, Florida, November 10-15, **2019**.

28. Yang Xiao, Anand Ramanathan, Bala Subramaniam and Arvind Varma, Enhanced Aromatic Selectivity during Deoxygenation of Phenolic Model Compounds over Bifunctional Catalysts, *2019 North American Catalysis Society Meeting (NAM26)*, Chicago, Illinois, June 23-28, **2019**.
27. Johnny ZhuChen, Zhenwei Wu, Slgi Choi, Yang Xiao, Arvind Varma, Wei Liu, Guanghui Zhang and Jeffrey T. Miller, Structure Determination of Nanoscale Pt_3Bi Intermetallic Alloy Catalysts for Non-Oxidative Coupling of Methane and Propane Dehydrogenation, *2019 North American Catalysis Society Meeting (NAM26)*, Chicago, Illinois, June 23-28, **2019**.
26. Yang Xiao and Arvind Varma, A Mechanistic Study of Glycerol Conversion to Aromatic Hydrocarbons over Bifunctional Metal-Supported H-ZSM-5 Catalysts, *The 4th North American Symposium for Chemical Reaction Engineering (NASCRE 4)*, ***invited talk in honor of Prof. Dan Luss' contributions to the field of chemical reaction engineering***, Houston, Texas, March 10-13, **2019**.
25. Yang Xiao and Arvind Varma, Highly Selective Nonoxidative Coupling of Methane (NOCM) over Pt-Bi Bimetallic Catalysts, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
24. Arvind Varma and Yang Xiao, Bio-Oil Upgrading Using Methane: A Mechanistic Study of Model Compound Guaiacol Reactions over Pt-Bi Bimetallic Catalysts, ***invited talk in honor of Prof. Doraiswami Ramkrishna' contributions to the field of chemical reaction engineering***, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
23. Yang Xiao, Yuan Wang and Arvind Varma, Methanol Conversion to Formaldehyde at Low Temperatures over Pt-Bi Bimetallic Catalysts, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
22. Yang Xiao and Arvind Varma, Bimetallic Catalysis for Various Shale Gas and Biomass Conversions, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
21. Rexonni B. Lagare, Yang Xiao and Arvind Varma, Direct Catalytic Conversion of Methane to Liquid Oxygenates *CISTAR Biannual Meeting*, Albuquerque, NM, October 3-5, **2018**.
20. Yang Xiao and Arvind Varma, Upgrading Bio-Oil Model Compound over Pt-Based Catalysts: A Comparative Study Using Hydrogen and Methane, *International Symposium on Chemical Reaction Engineering (ISCRE-25)*, Florence, Italy, May 20 - 23, **2018**.
19. Yang Xiao and Arvind Varma, Highly Selective Nonoxidative Coupling of Methane (NOCM) over Pt-Bi Bimetallic Catalysts *The Catalysis Club of Chicago 2018 Spring Symposium*, Chicago, Illinois, May 11, **2018**.
18. Yang Xiao, Yuan Wang and Arvind Varma, Low-Temperature Selective Oxidation of Methanol to Formaldehyde over Pt-Bi Bimetallic Catalysts, *39th Annual Michigan Catalysis Society Spring Symposium*, Midland, Michigan, May 3, **2018**.
17. Lindsey Blanshan, Yang Xiao and Arvind Varma, Catalytic Conversion of Biomass Model Compounds to Biofuels over Pt-based Supported Catalysts, *AIChE North Central Regional Conference*, West Lafayette, IN, April 6 - 7, **2018**.
16. Yang Xiao and Arvind Varma, Insight and Applications of Pt-Bi Bimetallic Catalysts: A Combined Experimental and DFT Study, *AIChE Annual Meeting*, Minneapolis, MN, October 29 - November 3, **2017**.

15. Yang Xiao and Arvind Varma, Kinetics of Glycerol Conversion to Hydrocarbon Fuels over Pd/H-ZSM-5 Catalysts, *AIChE Annual Meeting*, Minneapolis, MN, October 29 - November 3, **2017**.
14. Yang Xiao and Arvind Varma, Kinetics of Guaiacol Deoxygenation Using Methane over Pt-Bi Catalyst, *5th North American Catalysis Society Meeting*, Denver, CO, June 4-9, **2017**.
13. Xiaohong Wang, Yang Xiao and Arvind Varma, Controllable Solution Combustion Synthesis of Nanoscale α / $\beta - Bi_2O_3$ and Its Catalytic Application, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
12. Yang Xiao and Arvind Varma, Glycerol to Hydrocarbon Fuels Via Bifunctional Catalysts, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
11. Yang Xiao and Arvind Varma, Guaiacol Deoxygenation Using Methane over Pt-Bi Catalysts: Reaction Pathways and Kinetics, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
10. Yang Xiao and Arvind Varma, Insight into Pt-Bi Bimetallic Catalysts: An Experimental and DFT Study, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
9. Yang Xiao and Arvind Varma, Insight into Pt-Bi Bimetallic Catalysts for Tuning Selectivity and Improving Stability, *International Symposium on Chemical Reaction Engineering (ISCRE-24)*, Minneapolis, MN, June 12-15, **2016**.
8. Yang Xiao and Arvind Varma, Catalytic Deoxygenation of Guaiacol Using Methane, *AIChE Annual Meeting*, Salt Lake City, UT, November 8-13, **2015**.
7. Yang Xiao, Zhi-Jian Zhao, Jeffrey Greeley and Arvind Varma, An Experimental and Theoretical Study of Glycerol Selective Oxidation to 1,3-Dihydroxyacetone Via Bimetallic Platinum-Bismuth Catalysts, *AIChE Annual Meeting*, Salt Lake City, UT, November 8-13, **2015**.
6. Yang Xiao, Zhi-Jian Zhao, Jeffrey Greeley and Arvind Varma, Glycerol Selective Oxidation to 1,3-Dihydroxyacetone via Bimetallic Platinum-Bismuth Catalysts: An Experimental and Theoretical Study, *AIChE 7th Annual Midwest Regional Conference*, Chicago, IL, March 12-13, **2015**.
5. Yang Xiao, Guomin Xiao and Arvind Varma, Experimental and Simulation Study of a Universal Procedure for Crude Glycerol Purification From Different Feedstocks in Biodiesel Production, *AIChE Annual Meeting*, San Francisco, CA, November 3-8, **2013**.
4. Yang Xiao and Arvind Varma, Experimental and Simulation Study of Crude Glycerol Purification from Different Feedstocks in Biodiesel Production, *246th ACS National Meeting*, Indianapolis, IN, September 8-12, **2013**.
3. Yang Xiao, Haoyang Li, Guomin Xiao, Lijing Gao and Xiaomei Pan, Simulation of the Catalytic Reactive Distillation Process for Biodiesel Production via Transesterification, *International Conference on Renewable Energy and Environmental Materials*, Beijing, China, May 21-23, **2012**.
2. Lijing Gao, Guomin Xiao and Yang Xiao, Study on Intrinsic Kinetics of Transesterification Reaction Catalyzed by Solid Base in Fixed Bed Reactor, *The 6th National Conference on Chemical Engineering and Biochemistry*, Changsha, China, October 29-31, **2010**.
1. Yang Xiao, Lijing Gao, Guomin Xiao and Jianhua Lv, Kinetics of Transesterification Reaction Catalyzed by Solid Base in Fixed Bed Reactor, *International Conference on Chemical and Biological Utilization of Biomass Resources (ICCUB2010)*, Nanjing, China, October 22-26, **2010**.

Professional Services

Journal Editorships and Advisory Boards

Catalysts	Guest Editor
Catalysts	Topical Advisory Panel Member
Sustainability	Editorial Board Member

Proposal Reviewer

NSF	CBET Catalysis Program
ACS Petroleum Research Fund	Surface Science
DOE	Office of Basic Energy Sciences

Conference Sessions Chaired

5. Catalysis on Low Dimensional Materials, *AIChE Annual Meeting*, Orlando, FL, November 5 - 10, 2023.
4. Session III, *Southeastern Catalysis Society 2023 Annual Symposium*, Tuscaloosa, AL, February 27 - 28, 2023.
3. Fundamentals of Catalysis and Surface Science II: Zeolites and Acid Catalysis, *AIChE Annual Meeting*, Phoenix, AZ, November 13 - 18, 2022.
2. Biomass Upgrading I: Reaction Fundamentals, *AIChE Annual Meeting*, Boston, MA, November 5 - 11, 2021.
1. Fundamentals and Strategies for Catalytic Biomass Conversion, *AIChE Annual Meeting*, Virtual, November 16 - 20, 2020.

Manuscript Reviewer

ACS Omega, *ACS Sustainable Chemistry & Engineering*, *AIChE Journal*, *Applied Energy*, *Applied Catalysis A: General*, *Applied Surface Science*, *Asia-Pacific Journal of Chemical Engineering*, *Biomass and Bioenergy*, *Biomass Conversion and Biorefinery*, *Catalysis Communications*, *Catalysis Letters*, *Catalysts*, *Chem*, *Chem Catalysis*, *Chemical Engineering and Processing: Process Intensification*, *Chemical Engineering Journal*, *Chemical Engineering Science*, *ChemistrySelect*, *ChemSusChem*, *Chinese Journal of Chemical Engineering*, *ECS Journal of Solid State Science and Technology*, *Energy Exploration & Exploitation*, *Energy & Fuels*, *Fuel Processing*, *Journal of Environmental Chemical Engineering*, *Journal of Sustainable Bioenergy Systems*, *Industrial & Engineering Chemistry Research*, *Korean Journal of Chemical Engineering*, *Materials*, *Molecular Simulation*, *Molecular Physics*, *Molecular Catalysis*, *Molecules*, *Reaction Chemistry & Engineering*, *Reviews in Chemical Engineering*, *Sustainability*, and *Sustainable Energy & Fuels* .